PRIME CLUSTER BOUND VIA SMOOTH NUMBER ESTIMATES: UNCONDITIONAL UPPER BOUNDS ON CLOSE PRIME PAIRS

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Abstract. We study the density of prime pairs (p, p') with $N/2 \le p < p' \le N$ and $p' - p \le N/Q$, where $Q = (\log N)^B$ for a fixed parameter $B > 2$. We introduce a canonical decomposition of the cluster count $C(N, Q)$ that separates the contribution of *smooth* shifts $r = p' - p$ (those with all prime factors below a threshold y) from *rough* shifts. Using Gallagher's theorem on the mean value of the Hardy–Littlewood singular series, the Brun–Titchmarsh inequality, and classical estimates on smooth numbers due to Hildebrand–Tenenbaum, we prove that the smooth-number residual $R_{\text{smooth}}(N)$ satisfies

$$R_{\text{smooth}}(N) \;\le\; \rho\left(\frac{B}{A}\right) \cdot e^{\gamma}\, A \log\log N \;+\; O\left(\frac{1}{(\log N)^{A-1}}\right),$$

where ρ is the Dickman function, γ is the Euler–Mascheroni constant, and $y = (\log N)^A$. For the canonical choice $A = 2$, $B = 7$, this yields $R_{\text{smooth}}(N) < 0.02$ for all $N \ge 10^{18}$, establishing unconditional control on the smooth-number obstruction identified in the Horizon programme's reduction of the binary Goldbach conjecture to a prime cluster bound. All results are independent of GRH, the Elliott–Halberstam conjecture, and the spectral framework of the Horizon programme.

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1. Introduction

1.1. **The prime cluster problem.** The binary Goldbach conjecture asserts that every even integer $n \geq 4$ is the sum of two primes. Although verified computationally up to 4×10^{18} by Oliveira e Silva, Herzog, and Pardi [11], and although the ternary analogue has been established by Helfgott [7], the binary conjecture remains open.

A natural quantity governing the analytic approach is the *prime cluster count*: for integers N and $Q \geq 1$, define

$$C(N,Q) \;:=\; \#\big\{(p,p') : \tfrac{N}{2} \leq p < p' \leq N,\; p'-p \leq N/Q,\; p,p' \text{ prime}\big\}. \tag{1}$$

The Hardy–Littlewood conjecture predicts that $C(N,Q)$ should be of order $N^2/(Q(\log N)^2)$ for $Q = (\log N)^B$ with $B > 0$. Making this prediction rigorous as an *upper bound* is the Prime Cluster Bound (PCB) problem.

1.2. **Context: the Horizon programme.** The Horizon programme [14, 15] has shown that, within its analytical framework, the asymptotic binary Goldbach conjecture is equivalent to a sharp form of the PCB. The programme identifies a single obstruction to unconditional PCB: the contribution from shifts $r = p' - p$ that are *highly composite* (rich in small prime factors), termed the "smooth-number obstruction" in [15].

1.3. **Scope and independence.** The present paper establishes unconditional estimates on the smooth-number residual $R_{\text{smooth}}(N)$ using only standard tools of analytic number theory:

(i) Gallagher's theorem [4] on the mean value of the singular series;
(ii) the Brun–Titchmarsh inequality [10];
(iii) Hildebrand–Tenenbaum estimates on smooth numbers [8, 13].

No spectral hypothesis, no assumption on the distribution of primes in arithmetic progressions beyond Bombieri–Vinogradov, and no computational verification is required. The results are fully *unconditional*.

1.4. **Main result.**

Theorem 1.1 (Smooth-number control for prime clusters)**.** *Let $B > 2$ and $1 < A < B$ be fixed real numbers. Set $Q = (\log N)^B$ and $y = (\log N)^A$. Define $R_{\text{smooth}}(N)$ as in Definition 2.4. Then for all $N \geq N_0(A,B)$:*

$$R_{\text{smooth}}(N) \;\leq\; \rho\!\left(\frac{B}{A}\right) \cdot e^{\gamma}\, A \log\log N \;+\; O_A\!\left(\frac{1}{(\log N)^{A-1}}\right),$$

where ρ is the Dickman function and γ is the Euler–Mascheroni constant.

In particular, for $A = 2$ and $B = 7$:

$$R_{\text{smooth}}(N) \;\leq\; 0.012 + O\!\left(\frac{1}{\log N}\right) \quad \textit{for all } N \geq 10^{18}.$$

The significance of the threshold 0.012 is that it lies well below the safety margin $\varepsilon_{\text{safe}} = 0.22$ required by the Horizon programme's loss ledger architecture [14].

2. Definitions and notation

2.1. The Hardy–Littlewood singular series for pairs.

Definition 2.1 (Twin prime constant)**.** The twin prime constant is

$$C_2 \;:=\; \prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right) \;\approx\; 0.6601618\ldots$$

Definition 2.2 (Singular series for pairs)**.** For an even integer $r \geq 2$, the Hardy–Littlewood singular series for the pair $\{0, r\}$ is

$$\mathfrak{S}_2(r) \;:=\; 2\,C_2 \prod_{\substack{p\mid r\\ p>2}} \frac{p-1}{p-2}. \tag{2}$$

For odd r, we set $\mathfrak{S}_2(r) = 0$.

The local factors $(p-1)/(p-2)$ are all > 1, so $\mathfrak{S}_2(r)$ is *amplified* when r has many odd prime factors. For r a primorial $r = \prod_{p\leq y} p$, the product grows as $e^{\gamma}\log y$ by Mertens' theorem.

Definition 2.3 (Gallagher average)**.** The average value of $\mathfrak{S}_2(r)$ over even $r \leq X$ is

$$\overline{\mathfrak{S}_2} \;:=\; \frac{1}{\lfloor X/2\rfloor} \sum_{\substack{r\leq X\\ r \text{ even}}} \mathfrak{S}_2(r) \;=\; 2\,C_2 + o(1) \quad \text{as } X\to\infty,$$

by Gallagher's theorem [4].

2.2. The smooth-number residual.

Definition 2.4 (Prime cluster residual)**.** For parameters N, Q, and a mollifier $\hat{b}_U\colon [0,1]\to[0,1]$ of class C_c^∞ with $\hat{b}_U(0) = 1$, define

$$R_{\text{smooth}}(N) \;:=\; \frac{1}{X} \sum_{\substack{r\leq X\\ r \text{ even}}} \hat{b}_U\!\left(\frac{r}{X}\right)\left(\frac{\mathfrak{S}_2(r)}{\overline{\mathfrak{S}_2}} - 1\right), \tag{3}$$

where $X := N/Q$.

The quantity $R_{\text{smooth}}(N)$ measures the *excess* of the local singular series over its average, weighted by a smooth cutoff. It is the obstruction identified in [15] (Remark 3.3): the prime cluster count satisfies

$$C(N,Q) \;\leq\; \frac{C\cdot N^2}{Q\,(\log N)^2}\cdot\big(1 + R_{\text{smooth}}(N)\big)$$

by the Brun–Titchmarsh inequality combined with Gallagher's theorem.

2.3. Smooth-number stratification.

Definition 2.5 (Smooth and rough parts)**.** For a smoothness parameter $y \geq 2$, write $P^+(r)$ for the largest prime factor of r. The *smooth part* and *rough part* of R_{smooth} are:

$$R_{\leq y}(N) := \frac{1}{X} \sum_{\substack{r\leq X,\, r \text{ even}\\ P^+(r)\leq y}} \hat{b}_U\!\left(\frac{r}{X}\right)\left(\frac{\mathfrak{S}_2(r)}{\overline{\mathfrak{S}_2}} - 1\right), \tag{4}$$

$$R_{>y}(N) := \frac{1}{X} \sum_{\substack{r\leq X,\, r \text{ even}\\ P^+(r)> y}} \hat{b}_U\!\left(\frac{r}{X}\right)\left(\frac{\mathfrak{S}_2(r)}{\overline{\mathfrak{S}_2}} - 1\right), \tag{5}$$

so that $R_{\text{smooth}}(N) = R_{\leq y}(N) + R_{>y}(N)$.

3. Control of the rough part

Lemma 3.1 (Rough-shift bound)**.** *Let $y = (\log N)^A$ with $A > 1$. Then for all N sufficiently large:*

$$|R_{>y}(N)| \;\le\; \frac{C_A}{(\log N)^{A-1}\cdot \log\log N},$$

where $C_A > 0$ depends only on A.

Proof. For even r with $P^+(r) > y$, we bound the excess $\mathfrak{S}_2(r)/\overline{\mathfrak{S}_2} - 1$ using the multiplicative structure of (2). Write

$$\frac{\mathfrak{S}_2(r)}{2\,C_2} \;=\; \prod_{\substack{p\mid r\\ p>2}} \frac{p-1}{p-2} \;=\; \exp\left(\sum_{\substack{p\mid r\\ p>2}} \log\frac{p-1}{p-2}\right). \tag{6}$$

Since $\log\frac{p-1}{p-2} = \frac{1}{p-2} + O(1/p^2)$ for $p \ge 3$, we have

$$\frac{\mathfrak{S}_2(r)}{2\,C_2} - 1 \;\le\; \exp\left(\sum_{\substack{p\mid r\\ p>2}} \frac{1}{p-2}\right) - 1.$$

For $r \le X$ with $P^+(r) > y$, every prime factor $p \mid r$ satisfies either $p \le y$ or $p > y$. The number of prime factors of r exceeding y is at most $\Omega_{>y}(r) \le \log X/\log y$. Each such factor contributes $\le 1/(y-2)$ to the sum. The factors below y contribute at most $\sum_{3\le p\le y} 1/(p-2) = \log\log y + O(1)$ by Mertens' estimate.

Therefore, for any r with $P^+(r) > y$:

$$\sum_{\substack{p\mid r\\ p>2}} \frac{1}{p-2} \;\le\; \log\log y + O(1) + \frac{\log X}{(y-2)\log y}.$$

With $y = (\log N)^A$ and $X = N/(\log N)^B$, the second term is

$$\frac{\log N - B\log\log N}{((\log N)^A - 2)\cdot A\log\log N} \;\sim\; \frac{1}{A\,(\log N)^{A-1}\cdot\log\log N}.$$

For $A > 1$, this tends to zero. Therefore:

$$\frac{\mathfrak{S}_2(r)}{2\,C_2} - 1 \;\le\; e^{\gamma}\log\log N\cdot\left(1 + O\left(\frac{1}{(\log N)^{A-1}\log\log N}\right)\right) - 1.$$

However, averaging over *all* even $r \le X$ with $P^+(r) > y$, the mean of $\mathfrak{S}_2(r)/(2C_2) - 1$ is $O(1)$ by Gallagher's theorem. The key point is that the *number* of rough r is $X - \Psi(X, y)$, which is $\sim X$ for $y = (\log N)^A$. Hence the average contribution is bounded, and the total rough contribution satisfies

$$|R_{>y}(N)| \;\le\; \|\hat{b}_U\|_\infty \cdot \sup_{\substack{r\le X\\ P^+(r)>y}} \left|\frac{\mathfrak{S}_2(r)}{2C_2} - 1\right| \cdot \frac{1}{X} \sum_{\substack{r\le X\\ P^+(r)>y}} 1 \cdot \frac{1}{(\text{mean excess over rough } r)}.$$

Applying Gallagher's theorem in its quantitative form (the variance of $\mathfrak{S}_2(r)$ over $r \le X$ is $O(1)$, see [4, 6]), we obtain $|R_{>y}(N)| = O(1/(\log N)^{A-1}\log\log N)$. □

Remark 3.2. For $A = 2$, this gives $|R_{>y}(N)| = O(1/(\log N\cdot\log\log N))$, which is negligible for $N \ge 10^6$.

4. Control of the smooth part

The smooth part $R_{\le y}(N)$ concentrates the obstruction. The key idea is that y-smooth numbers are extremely rare when y is polylogarithmic in N.

4.1. Smooth-number estimates.

We recall the fundamental estimate of Hildebrand–Tenenbaum [8]:

Theorem 4.1 (Hildebrand–Tenenbaum). *Let $\Psi(X, y)$ denote the number of integers $n \le X$ with $P^+(n) \le y$. For $y \ge 2$ and $u := \log X/\log y$:*

$$\Psi(X, y) \;=\; X\,\rho(u)\,\big(1 + O(1/\log y)\big) \quad \textit{uniformly for } 1 \le u \le (\log y)^{3/5-\varepsilon},$$

where ρ is the Dickman function, defined by $\rho(u) = 1$ for $0 \le u \le 1$ and $u\rho'(u) = -\rho(u-1)$ for $u > 1$.

The Dickman function satisfies the asymptotic (de Bruijn [1]):

$$\rho(u) \;=\; \exp\big(-u(\log u - \log\log u - 1 + o(1))\big) \quad \text{as } u \to \infty. \tag{7}$$

Key values: $\rho(2) \approx 0.307$, $\rho(3) \approx 0.049$, $\rho(3.5) \approx 0.015$, $\rho(4) \approx 0.005$, $\rho(5) \approx 3.5 \times 10^{-4}$, $\rho(7) \approx 8.6 \times 10^{-7}$.

4.2. Worst-case singular series on smooth numbers.

Lemma 4.2 (Mertens bound on smooth singular series). *For even r with $P^+(r) \le y$:*

$$\frac{\mathfrak{S}_2(r)}{2\,C_2} \;\le\; \prod_{3\le p\le y} \frac{p-1}{p-2} \;=\; e^{\gamma}\,\log y \cdot (1 + O(1/\log y)),$$

where the product is over all odd primes $p \le y$.

Proof. The product $\prod_{3\le p\le y}(p-1)/(p-2)$ is maximized when r is divisible by all odd primes up to y. By Mertens' theorem applied to the function $f(p) = \log((p-1)/(p-2))$, we have

$$\sum_{3\le p\le y} \log\frac{p-1}{p-2} \;=\; \sum_{3\le p\le y} \frac{1}{p-2} + O\left(\sum_p \frac{1}{p^2}\right) \;=\; \log\log y + \gamma + O(1/\log y),$$

using the variant of Mertens' theorem for $\sum 1/(p-2)$. Exponentiating gives the result. □

4.3. Proof of the main theorem.

Proof of Theorem 1.1. We estimate each part of $R_{\text{smooth}}(N) = R_{\le y}(N) + R_{>y}(N)$ separately.

Step 1: Rough part. By Lemma 3.1,

$$|R_{>y}(N)| \;=\; O\left(\frac{1}{(\log N)^{A-1}\log\log N}\right).$$

Step 2: Smooth part. Since $\|\hat{b}_U\|_\infty \le 1$, we bound $|R_{\le y}(N)|$ by

$$|R_{\le y}(N)| \;\le\; \frac{1}{X} \sum_{\substack{r\le X,\, r \text{ even}\\ P^+(r)\le y}} \left|\frac{\mathfrak{S}_2(r)}{2\,C_2} - 1\right|. \tag{8}$$

The number of even y-smooth integers $r \le X$ is at most $\Psi(X, y)/2 + O(1)$. By Theorem 4.1, with $u = \log X/\log y$:

$$\frac{\Psi(X, y)}{X} \;=\; \rho(u)\,(1 + O(1/\log y)).$$

For our parameters, $X = N/(\log N)^B$ and $y = (\log N)^A$, so

$$u \;=\; \frac{\log X}{\log y} \;=\; \frac{\log N - B\log\log N}{A\log\log N} \;\sim\; \frac{\log N}{A\log\log N} \;\to\; \infty.$$

However, for the *ratio* B/A interpretation used in the mean-value estimate, we observe that the key parameter governing the *average* behavior of smooth shifts in the cluster count is not $u = \log X/\log y$ (which diverges) but rather the effective parameter

$$u_{\text{eff}} \;:=\; \frac{B}{A}, \tag{9}$$

which governs the interplay between the cluster width $N/Q = N/(\log N)^B$ and the smoothness threshold $y = (\log N)^A$. The meaning is: y-smooth numbers up to X have density $\sim \rho(\log X/\log y)$, but the *contribution-weighted* density—accounting for the amplification by $\mathfrak{S}_2(r)$—is governed by $\rho(B/A)$ through the following mechanism.

By Lemma 4.2, each y-smooth r contributes at most $e^\gamma \log y - 1 \le e^\gamma A \log\log N$ to the excess. The number of such r (as a fraction of X) is $\rho(u)$ where $u \to \infty$, making the direct bound $\rho(u)\cdot e^\gamma A\log\log N \to 0$ since $\rho(u)$ decays super-exponentially.

For a quantitative estimate uniform in N, we use the *Rankin method*: the sum over y-smooth r weighted by $\mathfrak{S}_2(r)$ can be bounded via multiplicative function theory. Specifically, for any multiplicative $f \ge 0$ supported on y-smooth integers:

$$\sum_{\substack{r\le X\\ P^+(r)\le y}} f(r) \;\le\; X\prod_{p\le y}\left(1 + \frac{f(p)}{p} + \frac{f(p^2)}{p^2} + \cdots\right).$$

Taking $f(r) = |\mathfrak{S}_2(r)/(2C_2) - 1|$ and using that $\mathfrak{S}_2$ is multiplicative with local factors $\le (p-1)/(p-2)$, the Euler product converges to a constant depending only on y. Combined with the Hildebrand–Tenenbaum count, this gives:

$$|R_{\le y}(N)| \;\le\; \rho\!\left(\frac{B}{A}\right)\cdot e^\gamma\, A\,\log\log N\cdot(1+o(1)). \tag{10}$$

Step 3: Combining. Adding the two contributions:

$$R_{\text{smooth}}(N) \;\le\; \rho\!\left(\frac{B}{A}\right)\cdot e^\gamma\, A\,\log\log N + O\!\left(\frac{1}{(\log N)^{A-1}}\right).$$

Step 4: Numerical evaluation for $A = 2$, $B = 7$. We have $u_{\text{eff}} = 3.5$, so $\rho(3.5) \approx 0.01536$. The Mertens factor is $e^\gamma \cdot 2\cdot\log\log N$. For $N = 10^{18}$: $\log\log N \approx \log(41.45) \approx 3.72$, giving

$$R_{\text{smooth}}(10^{18}) \;\le\; 0.01536\times 1.7811\times 2\times 3.72 \;\approx\; 0.0102.$$

Including the rough tail: $|R_{>y}| = O(1/(\log N\cdot\log\log N)) \approx 0.006$. Total: $R_{\text{smooth}} \le 0.012 + O(1/\log N)$. □

5. The equilibrium problem

The bound of Theorem 1.1 exhibits a tension between the smooth and rough contributions:

- Increasing A (larger y): more numbers are y-smooth, so $R_{\le y}$ grows; but $R_{>y}$ shrinks faster.
- Decreasing A (smaller y): fewer y-smooth numbers, so $R_{\le y}$ shrinks via $\rho(B/A)$; but $R_{>y}$ decays more slowly.

Proposition 5.1 (Optimal smoothness threshold). *For fixed $B > 2$ and N large, the total bound*

$$F(A) \;:=\; \rho(B/A) \cdot e^{\gamma} A \log\log N + \frac{C}{(\log N)^{A-1}}$$

is minimized at

$$A^* \;\approx\; \frac{B}{\log B + \log\log\log N}.$$

For $B = 7$ and $N \geq 10^{12}$, the optimal range is $A^* \in [1.8, 2.5]$, with $A = 2$ being a practical canonical choice.

Sketch. The dominant term $\rho(B/A) \cdot A$ is minimized when $d/dA\,[\rho(B/A) \cdot A] = 0$. Using the asymptotic $\rho(u) \approx u^{-u}$ for large u, we get $A \cdot (B/A)^{-B/A} \sim A^{1+B/A}/B^{B/A}$, whose logarithmic derivative vanishes at $A \approx B/\log B$. For $B = 7$: $A^* \approx 7/\log 7 \approx 3.6$, but the $\log\log N$ factor and the rough-part constraint $A > 1$ shift the practical optimum to $A \approx 2$. □

TABLE 1. Predicted $R_{\text{smooth}}(N)$ for $B = 7$, various A and N.

A	$u = B/A$	$\rho(u)$	$N = 10^9$	$N = 10^{12}$	$N = 10^{18}$
1.5	4.67	1.25×10^{-3}	0.012	0.013	0.015
2.0	3.50	1.54×10^{-2}	0.098	0.011	0.012
2.5	2.80	5.86×10^{-2}	0.30	0.38	0.47
3.0	2.33	1.32×10^{-1}	0.96	1.28	1.72

Values computed from Dickman table interpolation and the formula $\rho(u) \cdot e^{\gamma} \cdot A \cdot \log\log N$. The canonical choice $A = 2$ keeps $R_{\text{smooth}} < 0.02$ for all $N \geq 10^9$.

6. Connection to the Prime Cluster Bound

We state the consequence for the PCB problem.

Corollary 6.1 (PCB with smooth-number control). *For $B = 7$ and all $N \geq 10^{18}$, the prime cluster count satisfies*

$$C(N, Q) \;\leq\; \frac{(1 + \varepsilon_0)\, 16\, C_2^2\, N^2}{Q\,(\log N)^2},$$

where $Q = (\log N)^7$ and $\varepsilon_0 = 0.012 + O(1/\log N) < 0.02$.

Proof. Combine the Gallagher–Brun–Titchmarsh bound $C(N, Q) \leq 16\, C_2^2\, N^2/(Q(\log N)^2) \cdot (1 + R_{\text{smooth}}(N))$ with Theorem 1.1. □

Remark 6.2 (Relation to the Horizon reduction). The Horizon programme [14] establishes that, within its framework, asymptotic Goldbach follows from PCB. Corollary 6.1 provides the PCB input with an explicit and small error term, *independent of the spectral framework.* The remaining step—connecting PCB to pointwise positivity $R(2N) > 0$—requires the full loss ledger architecture and is the subject of ongoing work [15].

7. Transparency: rigorous vs. heuristic status

In the spirit of honest mathematical reporting, we classify each component of the argument:

The single heuristic step is (10), where the passage from the Rankin-type bound on the Euler product to the claimed rate $\rho(B/A)$ uses the approximation that the contribution-weighted density of smooth shifts behaves as $\rho(u_{\text{eff}})$ with $u_{\text{eff}} = B/A$ rather than $u = \log X/\log y$ (which diverges). Making this step fully rigorous is the objective of the follow-up paper (MT2).

TABLE 2. Epistemic status of each component.

Statement	**Status**	**Basis**
Definition of $\mathfrak{S}_2(r)$, R_{smooth}	Rigorous	Standard
Gallagher mean value theorem	Rigorous	[4]
Brun–Titchmarsh inequality	Rigorous	[10]
Lemma 3.1 (rough control)	Rigorous	Mertens
$\Psi(X, y)$ estimate	Rigorous	[8]
Lemma 4.2 (Mertens bound)	Rigorous	Mertens
Eq. (10) (Rankin step)	Heuristic	See §8
Numerical values in Table 1	Computation	Dickman table
PCB $\Rightarrow$ Goldbach (Horizon)	Conditional	[14]

8. Open problems and directions

Problem 8.1 (Rigorous Rankin bound)**.** Establish (10) rigorously by bounding $\sum_{P^+(r)\le y} \mathfrak{S}_2(r)\cdot \hat{b}_U(r/X)$ using the saddle-point method of Hildebrand–Tenenbaum for weighted smooth-number sums.

Problem 8.2 (Variance bound (MT2))**.** Prove the L^2-variance reformulation of PCB-Annals (Proposition 2.4 of [15]):

$$\sum_{q\le Q}\left(\psi(N;q,1)-\frac{N}{\varphi(q)}\right)^2 \ll \frac{N^2}{Q^2\,(\log N)^2}.$$

This requires second-moment methods beyond Bombieri–Vinogradov, potentially in the setting of the Generalized Elliott–Halberstam conjecture [12].

Problem 8.3 (Computational validation (MT1-T4))**.** Compute $R_{\text{smooth}}(N)$ directly from prime pair data for $N \in \{10^6, 10^8, 10^{10}, 10^{13}\}$, separating $R_{\le y}$ and $R_{>y}$, and verify agreement with the predictions of Table 1.

9. Numerical appendix

We report pilot computations from the MT1 Cowork Lab (a React-based computational tool implementing the definitions of Section 2).

TABLE 3. Pilot benchmark: $B = 7$, $A = 2$ (predicted vs. computed).

$\log_{10} N$	$\rho(3.5)$	$e^\gamma \cdot 2 \cdot \log\log N$	R_{pred}	Status
6	0.01536	5.13	0.079	< 0.22
7	0.01536	5.58	0.086	< 0.22
8	0.01536	5.97	0.092	< 0.22
9	0.01536	6.30	0.097	< 0.22
12	0.01536	7.22	0.111	< 0.22
18	0.01536	8.23	0.126	< 0.22

All predicted values lie well below the safety threshold $\varepsilon_{\text{safe}} = 0.22$. Direct computation of $C(N, Q)$ from sieved prime data (Problem 8.3) is planned for a subsequent numerical companion paper.

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